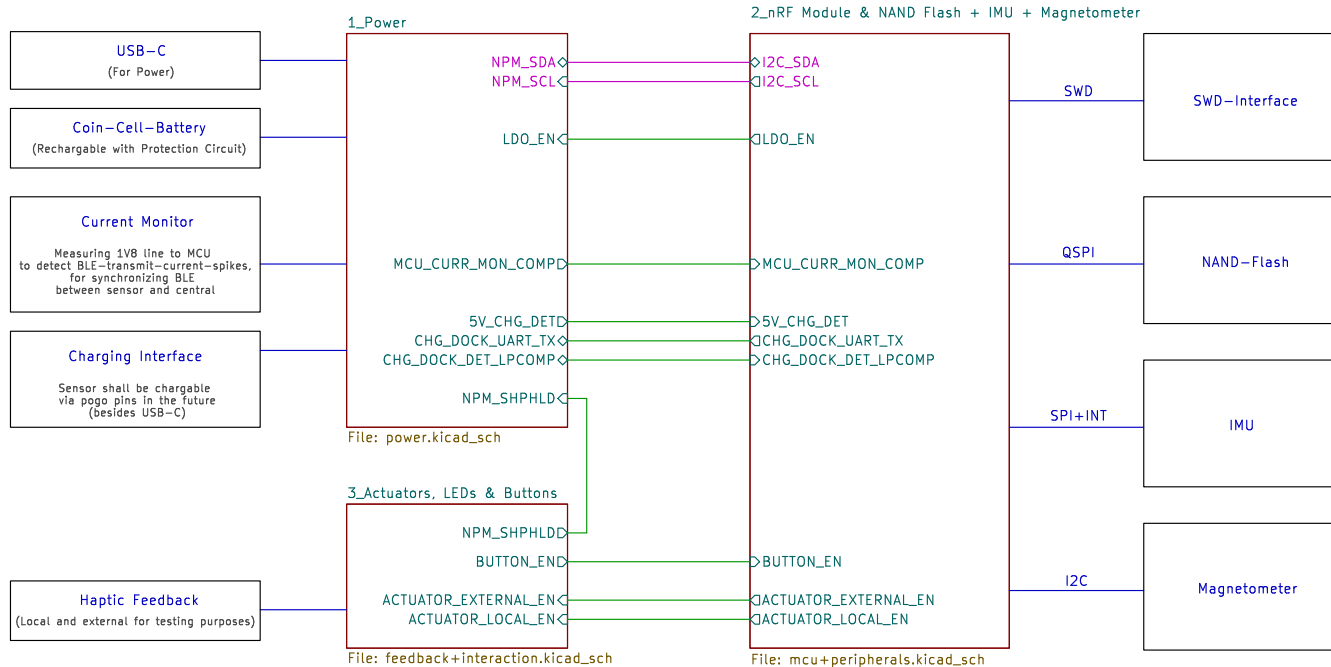


SENSOR – IEMU Development Board – Open-source Inertial Sensor for Biomechanics

Overview



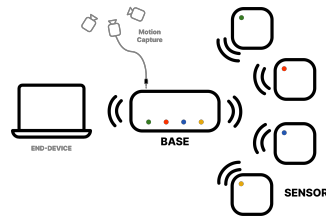
Test Points

TP101	+VBUS_IN
TP102	+VBAT
TP103	-VBAT
TP104	+VSY5
TP105	+3V0
TP106	+1V8_MCU
TP107	+1V8_PERIPHERALS
TP108	LDO_EN
TP109	MCU_CURR_MON
TP110	CHG_DOCK_DET_LPCOMP
TP111	CHG_DOCK_UART_TX
TP201	SWD_CLK
TP202	SWD_IO
TP203	SWD_RESET
TP204	QSPI_CS
TP205	SPI_CS_IMU
TP206	SPI_CLK_IMU
TP207	INT_IMU
TP208	I2C_SCL
TP209	I2C_SDA
TP301	ACTUATOR_LOCAL_EN
TP302	ACTUATOR_EXTERNAL_EN
TP303	NPM_SHPHLD
TP304	BUTTON_EN
TP112	GND
TP113	GND
TP114	GND

Concept

This is the development board for a sensor, that will be used for biomechanical measurements of acceleration and angular velocity (IMU). The magnetometer is added for hopefully improved sensor fusion. The NAND-Flash should allow about 2 hours of recording at ~1600Hz IMU + 100Hz Magnetometer.

The sensor will be reporting back to a central node (base). The base will be equipped with a SD-Card / more storage and will be the middle man communicating back and forth between the sensor and the users device (eg smartphone). Most importantly the central node will be synchronizing multiple sensors via its timestamp sent with the BLE-package. That's why the current monitor is measuring the current on the 1V8 line of the MCU, as it can detect the moment of ble transmit through a current spike. This can then be referred back to and matched with the timestamp sent from the central node. The central node will be serving as a charging case as well.



SENSOR – IEMU Development Board – Open-source Inertial Sensor for Biomechanics
Mustafa Acar

Sheet: /
File: IEMU_Sensor_DB.kicad_sch

Title: Project Overview

Size: A4 Date: 2026-04-03

KiCad E.D.A. 9.0.5

Rev:

Id: 1/4

POWER

CHARGING 4-PIN CONNECTOR

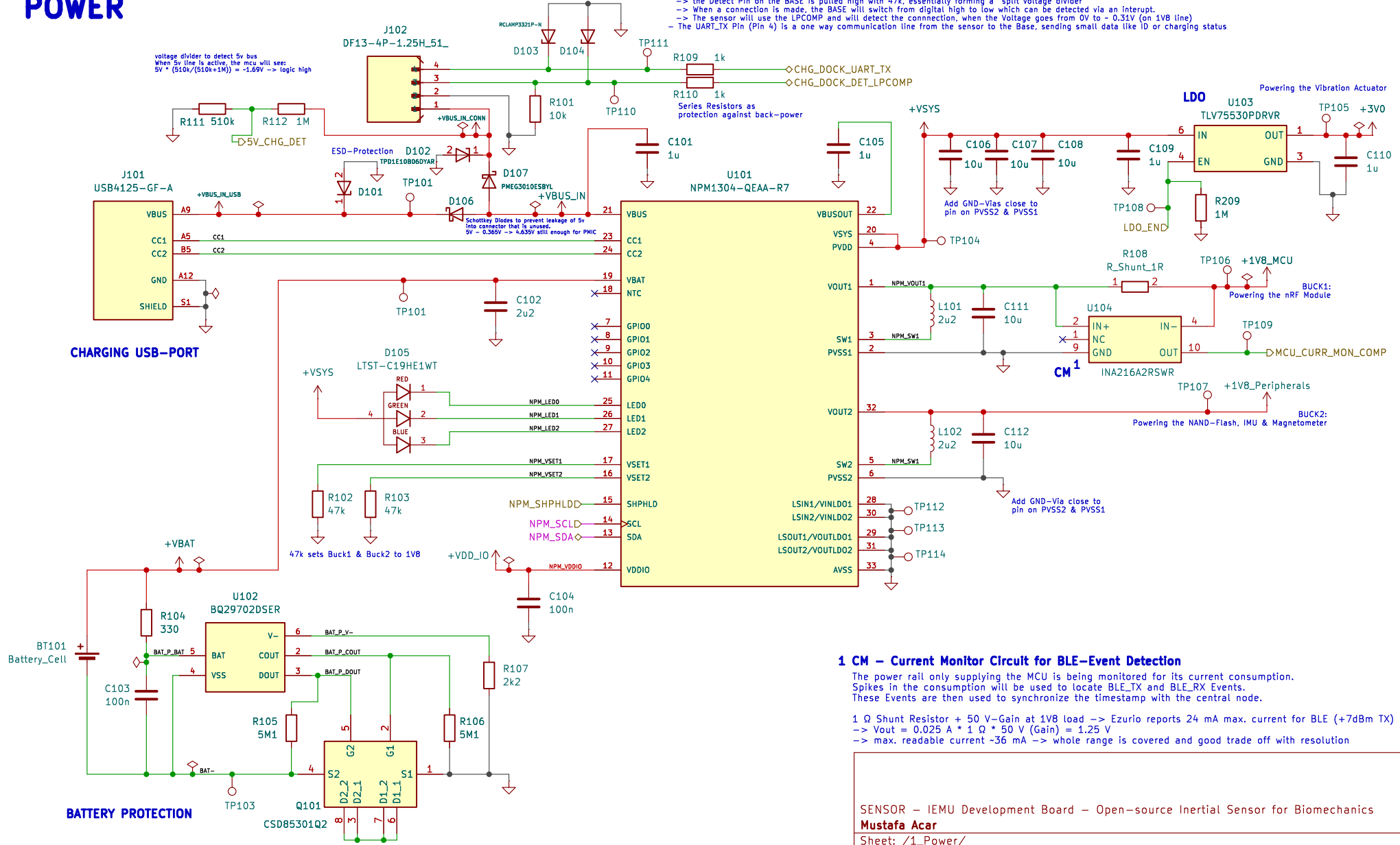
The connector is used to prototype a 4-pogo pin charging setup

- It just feeds the +VBUS with 5 V input (PIN 1)
- The Detect Pin (Pin 3) is pulled low with 10k resulting in a 0 V reading, when not connected.
- > the Detect Pin on the BASE is pulled high with 47k, essentially forming a "split voltage divider"
- > When a connection is made, the BASE will switch from digital high to low which can be detected via an interrupt.
- > The sensor will use the LPCOMP and will detect the connection, when the Voltage goes from 0V to ~ 0.31V (on 1V8 line)
- The UART_TX Pin (Pin 4) is a one way communication line from the sensor to the Base, sending small data like ID or charging status

voltage divider to detect 5v bus
When 5v line is active, the mcu will see:
 $5V \cdot (510k / (510k + 1M)) = \sim 1.69V \rightarrow$ logic high

CHARGING USB-PORT

BATTERY PROTECTION



1 CM – Current Monitor Circuit for BLE–Event Detection

The power rail only supplying the MCU is being monitored for its current consumption. Spikes in the consumption will be used to locate BLE_TX and BLE_RX Events. These Events are then used to synchronize the timestamp with the central node.

1 Ω Shunt Resistor + 50 V-Gain at 1V8 load -> Ezurio reports 24 mA max. current for BLE (+7dBm TX)
-> $V_{out} = 0.025 A \cdot 1 \Omega \cdot 50 V (Gain) = 1.25 V$
-> max. readable current ~36 mA -> whole range is covered and good trade off with resolution

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Sheet: /1_Power/
File: power.kicad_sch

Title: Power

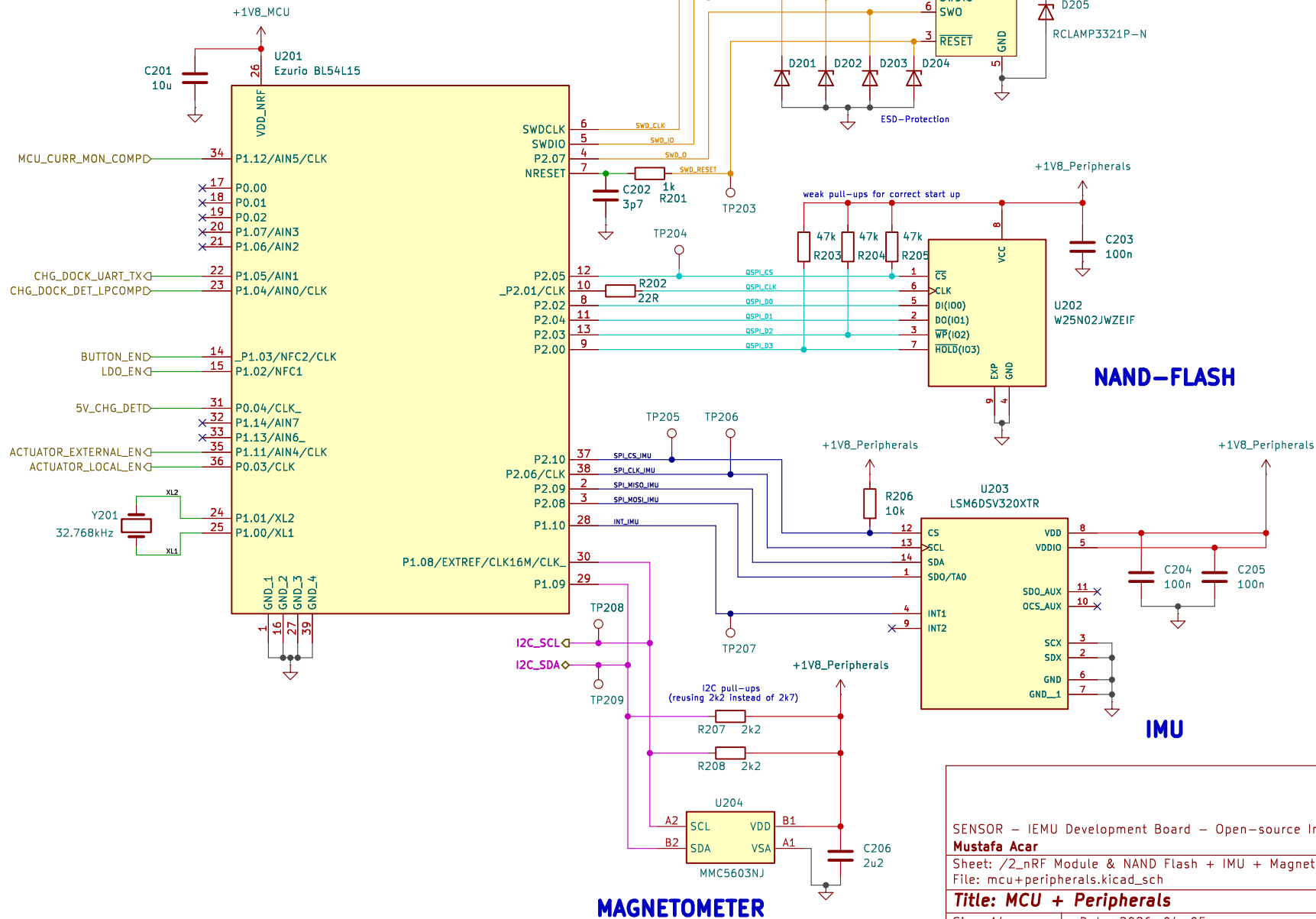
Size: A4 Date: 2026-04-03

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Id: 2/4

MCU & PERIPHERALS



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Sheet: /2_nRF Module & NAND Flash + IMU + Magnetometer/

File: mcu+peripherals.kicad_sch

Title: MCU + Peripherals

Size: A4 Date: 2026-04-05

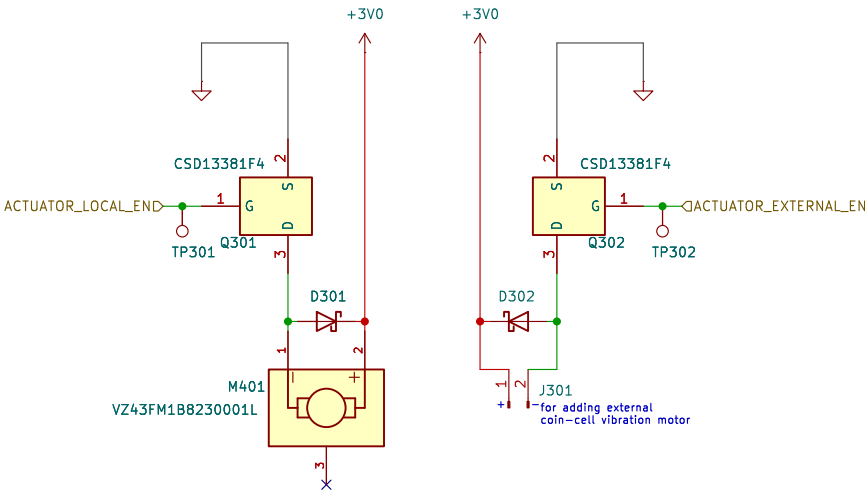
KiCad E.D.A. 9.0.5

Rev:

Id: 3/4

Actuators, LEDs & Buttons

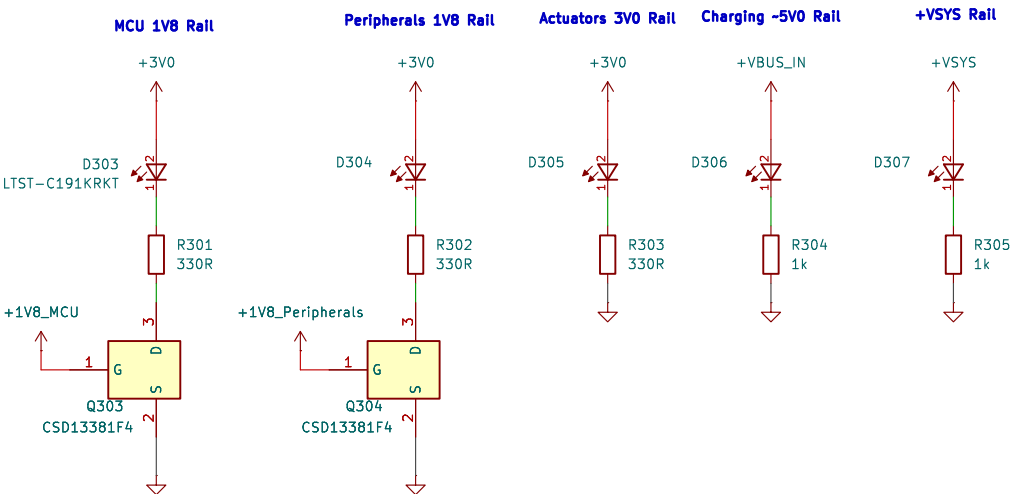
Haptic Actuators



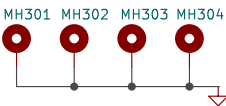
Buttons



Debug LEDs



Mounting Holes



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Sheet: /3_Actuators, LEDs & Buttons/
File: feedback+interaction.kicad_sch

Title: Actuators, LEDs & Buttons

Size: A4 Date: 2026-04-12

KiCad E.D.A. 9.0.5

Rev:
Id: 4/4